

Spectre Figure 5.1: Straight Positioned Join Audit

Immediate breakers first; compressed complete classification after

Straight-only catalogue: 76 positioned path records; 288 valid pre-image joins; 0 non-breaking inflated joins. There are 25 immediate breaks. Every immediate break has the same condition:

need $(\varepsilon^-, 1 : \varepsilon^+)$ available straight record: $\Phi_{4,3}$.

Immediate breakers (1 of 2)

Pre-image join	Post-image words	Positioned until immediate break
$\Delta_{1,3} \mid \Phi_{3,4}$	$(\Xi_4 \Delta_5 \Phi_0) (\Phi_4 \Sigma_5 \Phi_0)$	$(\Xi_{4,5} \Delta_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Sigma_5 \Phi_0)$
$\Delta_{1,3} \mid \Pi_{5,4}$	$(\Xi_4 \Delta_5 \Phi_0) (\Phi_4 \Gamma_2 \Psi_0)$	$(\Xi_{4,5} \Delta_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Gamma_2 \Psi_0)$
$\Delta_{1,3} \mid \Psi_{3,4}$	$(\Xi_4 \Delta_5 \Phi_0) (\Phi_4 \Sigma_5 \Phi_0)$	$(\Xi_{4,5} \Delta_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Sigma_5 \Phi_0)$
$\Delta_{1,3} \mid \Psi_{5,4}$	$(\Xi_4 \Delta_5 \Phi_0) (\Phi_4 \Gamma_2 \Psi_0)$	$(\Xi_{4,5} \Delta_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Gamma_2 \Psi_0)$
$\Delta_{1,3} \mid \Xi_{5,4}$	$(\Xi_4 \Delta_5 \Phi_0) (\Phi_4 \Gamma_2 \Psi_0)$	$(\Xi_{4,5} \Delta_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Gamma_2 \Psi_0)$
$\Gamma_{3,5} \mid \Delta_{3,3}$	$(\Theta_4 \Sigma_5 \Phi_0) (\Phi_4 \Sigma_5 \Phi_0)$	$(\Theta_{4,5} \Sigma_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Sigma_5 \Phi_0)$
$\Gamma_{3,5} \mid \Gamma_{5,3}$	$(\Theta_4 \Sigma_5 \Phi_0) (\Phi_4 \Gamma_2 \Pi_0)$	$(\Theta_{4,5} \Sigma_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Gamma_2 \Pi_0)$
$\Gamma_{3,5} \mid \Lambda_{3,3}$	$(\Theta_4 \Sigma_5 \Phi_0) (\Phi_4 \Sigma_5 \Phi_0)$	$(\Theta_{4,5} \Sigma_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Sigma_5 \Phi_0)$
$\Gamma_{3,5} \mid \Pi_{3,3}$	$(\Theta_4 \Sigma_5 \Phi_0) (\Phi_4 \Sigma_5 \Phi_0)$	$(\Theta_{4,5} \Sigma_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Sigma_5 \Phi_0)$
$\Gamma_{3,5} \mid \Sigma_{3,3}$	$(\Theta_4 \Sigma_5 \Phi_0) (\Phi_4 \Sigma_5 \Lambda_0)$	$(\Theta_{4,5} \Sigma_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Sigma_5 \Lambda_0)$
$\Pi_{3,3} \mid \Delta_{3,3}$	$(\Phi_4 \Sigma_5 \Phi_0) (\Phi_4 \Sigma_5 \Phi_0)$	$(\Phi_{4,3} \Sigma_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Sigma_5 \Phi_0)$
$\Pi_{3,3} \mid \Gamma_{5,3}$	$(\Phi_4 \Sigma_5 \Phi_0) (\Phi_4 \Gamma_2 \Pi_0)$	$(\Phi_{4,3} \Sigma_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Gamma_2 \Pi_0)$

Immediate breakers (2 of 2)

Pre-image join	Post-image words	Positioned until immediate break
$\Pi_{3,3} \mid \Lambda_{3,3}$	$(\Phi_4 \Sigma_5 \Phi_0) (\Phi_4 \Sigma_5 \Phi_0)$	$(\Phi_{4,3} \Sigma_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Sigma_5 \Phi_0)$
$\Pi_{3,3} \mid \Pi_{3,3}$	$(\Phi_4 \Sigma_5 \Phi_0) (\Phi_4 \Sigma_5 \Phi_0)$	$(\Phi_{4,3} \Sigma_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Sigma_5 \Phi_0)$
$\Pi_{3,3} \mid \Sigma_{3,3}$	$(\Phi_4 \Sigma_5 \Phi_0) (\Phi_4 \Sigma_5 \Lambda_0)$	$(\Phi_{4,3} \Sigma_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Sigma_5 \Lambda_0)$
$\Psi_{3,4} \mid \Delta_{3,3}$	$(\Phi_4 \Sigma_5 \Phi_0) (\Phi_4 \Sigma_5 \Phi_0)$	$(\Phi_{4,3} \Sigma_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Sigma_5 \Phi_0)$
$\Psi_{3,4} \mid \Gamma_{5,3}$	$(\Phi_4 \Sigma_5 \Phi_0) (\Phi_4 \Gamma_2 \Pi_0)$	$(\Phi_{4,3} \Sigma_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Gamma_2 \Pi_0)$
$\Psi_{3,4} \mid \Lambda_{3,3}$	$(\Phi_4 \Sigma_5 \Phi_0) (\Phi_4 \Sigma_5 \Phi_0)$	$(\Phi_{4,3} \Sigma_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Sigma_5 \Phi_0)$
$\Psi_{3,4} \mid \Pi_{3,3}$	$(\Phi_4 \Sigma_5 \Phi_0) (\Phi_4 \Sigma_5 \Phi_0)$	$(\Phi_{4,3} \Sigma_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Sigma_5 \Phi_0)$
$\Psi_{3,4} \mid \Sigma_{3,3}$	$(\Phi_4 \Sigma_5 \Phi_0) (\Phi_4 \Sigma_5 \Lambda_0)$	$(\Phi_{4,3} \Sigma_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Sigma_5 \Lambda_0)$
$\Sigma_{1,5} \mid \Phi_{3,4}$	$(\Xi_4 \Delta_5 \Phi_0) (\Phi_4 \Sigma_5 \Phi_0)$	$(\Xi_{4,5} \Delta_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Sigma_5 \Phi_0)$
$\Sigma_{1,5} \mid \Pi_{5,4}$	$(\Xi_4 \Delta_5 \Phi_0) (\Phi_4 \Gamma_2 \Psi_0)$	$(\Xi_{4,5} \Delta_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Gamma_2 \Psi_0)$
$\Sigma_{1,5} \mid \Psi_{3,4}$	$(\Xi_4 \Delta_5 \Phi_0) (\Phi_4 \Sigma_5 \Phi_0)$	$(\Xi_{4,5} \Delta_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Sigma_5 \Phi_0)$
$\Sigma_{1,5} \mid \Psi_{5,4}$	$(\Xi_4 \Delta_5 \Phi_0) (\Phi_4 \Gamma_2 \Psi_0)$	$(\Xi_{4,5} \Delta_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Gamma_2 \Psi_0)$
$\Sigma_{1,5} \mid \Xi_{5,4}$	$(\Xi_4 \Delta_5 \Phi_0) (\Phi_4 \Gamma_2 \Psi_0)$	$(\Xi_{4,5} \Delta_{5,2} \Phi_{0,3}) (\underline{\Phi_4} \Gamma_2 \Psi_0)$

Key for pages 3-5. J = break at the join between the parenthesized image words; L = break inside the left image word; R = break inside the right image word. The grouped tables cover all 288 rows; N gives the number of pre-image joins represented by each row.

Complete grouped classification (1 of 3)

#	N	Break	Post-image pattern	Required / available	Representative pre-images
1	6J	Φ_4	$(\Phi_4 \Sigma_5 \Phi_0) (\Phi_4 \Sigma_5 \Phi_0)$	<i>need</i> $(\varepsilon^-, 1 : \varepsilon^+)$; avail $\Phi_{4,3}$	$\Pi_{3,3} \mid \Delta_{3,3}; \Pi_{3,3} \mid \Lambda_{3,3}$
2	6J	Φ_4	$(\Xi_4 \Delta_5 \Phi_0) (\Phi_4 \Gamma_2 \Psi_0)$	<i>need</i> $(\varepsilon^-, 1 : \varepsilon^+)$; avail $\Phi_{4,3}$	$\Delta_{1,3} \mid \Pi_{5,4}; \Delta_{1,3} \mid \Psi_{5,4}$
3	4J	Φ_4	$(\Xi_4 \Delta_5 \Phi_0) (\Phi_4 \Sigma_5 \Phi_0)$	<i>need</i> $(\varepsilon^-, 1 : \varepsilon^+)$; avail $\Phi_{4,3}$	$\Delta_{1,3} \mid \Phi_{3,4}; \Delta_{1,3} \mid \Psi_{3,4}$
4	3J	Φ_4	$(\Theta_4 \Sigma_5 \Phi_0) (\Phi_4 \Sigma_5 \Phi_0)$	<i>need</i> $(\varepsilon^-, 1 : \varepsilon^+)$; avail $\Phi_{4,3}$	$\Gamma_{3,5} \mid \Delta_{3,3}; \Gamma_{3,5} \mid \Lambda_{3,3}$
5	2J	Φ_4	$(\Phi_4 \Sigma_5 \Phi_0) (\Phi_4 \Gamma_2 \Pi_0)$	<i>need</i> $(\varepsilon^-, 1 : \varepsilon^+)$; avail $\Phi_{4,3}$	$\Pi_{3,3} \mid \Gamma_{5,3}; \Psi_{3,4} \mid \Gamma_{5,3}$
6	2J	Φ_4	$(\Phi_4 \Sigma_5 \Phi_0) (\Phi_4 \Sigma_5 \Lambda_0)$	<i>need</i> $(\varepsilon^-, 1 : \varepsilon^+)$; avail $\Phi_{4,3}$	$\Pi_{3,3} \mid \Sigma_{3,3}; \Psi_{3,4} \mid \Sigma_{3,3}$
7	1J	Φ_4	$(\Theta_4 \Sigma_5 \Phi_0) (\Phi_4 \Gamma_2 \Pi_0)$	<i>need</i> $(\varepsilon^-, 1 : \varepsilon^+)$; avail $\Phi_{4,3}$	$\Gamma_{3,5} \mid \Gamma_{5,3}$
8	1J	Φ_4	$(\Theta_4 \Sigma_5 \Phi_0) (\Phi_4 \Sigma_5 \Lambda_0)$	<i>need</i> $(\varepsilon^-, 1 : \varepsilon^+)$; avail $\Phi_{4,3}$	$\Gamma_{3,5} \mid \Sigma_{3,3}$
9	3L	Γ_4	$(\Xi_3 \Delta_3 \Gamma_4) (\Xi_3 \Delta_3 \Gamma_4)$	<i>need</i> $(\gamma^-, 1 : \beta^+)$; avail $\Gamma_{4,3}$	$\Pi_{3,1} \mid \Delta_{3,1}; \Pi_{3,1} \mid \Lambda_{3,1}$
10	3L	Ξ_0	$(\Gamma_1 \Delta_0 \Xi_0) (\Gamma_1 \Delta_0 \Pi_0)$	<i>need</i> $(\alpha^-, 1 : \varepsilon^+)$; avail $\Xi_{0,3}$	$\Delta_{0,5} \mid \Xi_{0,3}; \Lambda_{0,5} \mid \Xi_{0,3}$
11	3L	Ξ_0	$(\Gamma_1 \Delta_0 \Xi_0) (\Gamma_1 \Delta_0 \Psi_0)$	<i>need</i> $(\alpha^-, 1 : \varepsilon^+)$; avail $\Xi_{0,3}$	$\Delta_{0,5} \mid \Psi_{0,3}; \Lambda_{0,5} \mid \Psi_{0,3}$
12	3L	Ξ_0	$(\Gamma_1 \Delta_0 \Xi_0) (\Gamma_1 \Delta_0 \Xi_0)$	<i>need</i> $(\alpha^-, 1 : \varepsilon^+)$; avail $\Xi_{0,3}$	$\Delta_{0,5} \mid \Pi_{0,3}; \Lambda_{0,5} \mid \Pi_{0,3}$
13	3L	Ξ_0	$(\Gamma_1 \Delta_0 \Xi_0) (\Gamma_1 \Delta_0)$	<i>need</i> $(\alpha^-, 1 : \varepsilon^+)$; avail $\Xi_{0,3}$	$\Delta_{0,5} \mid \Gamma_{0,3}; \Lambda_{0,5} \mid \Gamma_{0,3}$
14	6R	Δ_2	$(\Psi_3 \Gamma_5 \Phi_1) (\Phi_3 \Delta_2 \Xi_1)$	<i>need</i> $(\gamma^-, 1 : \beta^+)$; avail $\Delta_{2,3}$	$\Pi_{2,2} \mid \Delta_{4,1}; \Pi_{2,2} \mid \Sigma_{4,1}$
15	4R	Δ_2	$(\Phi_3 \Sigma_2 \Phi_1) (\Phi_3 \Delta_2 \Xi_1)$	<i>need</i> $(\gamma^-, 1 : \beta^+)$; avail $\Delta_{2,3}$	$\Phi_{0,3} \mid \Delta_{4,1}; \Phi_{0,3} \mid \Sigma_{4,1}$
16	18R	Γ_3	$(\Delta_4 \Sigma_0 \Psi_0) (\Sigma_4 \Gamma_3)$	<i>need</i> $(\delta^-, 1 : \theta^-)$; avail $\Gamma_{3,3}, \Gamma_{3,5}$	$\Pi_{2,4} \mid \Delta_{4,3}; \Pi_{2,4} \mid \Lambda_{4,3}$

Complete grouped classification (2 of 3)

#	N	Break	Post-image pattern	Required / available	Representative pre-images
17	12R	Γ_3	$(\Gamma_1 \Delta_0 \Psi_0) (\Sigma_4 \Gamma_3)$	<i>need</i> $(\delta^-, 1 : \theta^-)$; avail $\Gamma_{3,3}, \Gamma_{3,5}$	$\Phi_{0,5} \mid \Delta_{4,3}; \Phi_{0,5} \mid \Lambda_{4,3}$
18	3R	Γ_3	$(\Phi_4 \Sigma_5 \Phi_0) (\Sigma_4 \Gamma_3)$	<i>need</i> $(\delta^-, 1 : \theta^-)$; avail $\Gamma_{3,3}, \Gamma_{3,5}$	$\Delta_{3,3} \mid \Gamma_{4,3}; \Lambda_{3,3} \mid \Gamma_{4,3}$
19	2R	Γ_3	$(\Phi_4 \Gamma_2 \Psi_0) (\Sigma_4 \Gamma_3)$	<i>need</i> $(\delta^-, 1 : \theta^-)$; avail $\Gamma_{3,3}, \Gamma_{3,5}$	$\Pi_{5,4} \mid \Theta_{4,5}; \Pi_{5,4} \mid \Xi_{4,5}$
20	14R	Γ_4	$(\Sigma_4 \Gamma_3) (\Pi_3 \Delta_3 \Gamma_4)$	<i>need</i> $(\gamma^-, 1 : \beta^+)$; avail $\Gamma_{4,3}$	$\Gamma_{4,3} \mid \Theta_{3,2}; \Gamma_{4,3} \mid \Xi_{3,2}$
21	12R	Γ_4	$(\Gamma_0 \Sigma_1) (\Psi_3 \Delta_3 \Gamma_4)$	<i>need</i> $(\gamma^-, 1 : \beta^+)$; avail $\Gamma_{4,3}$	$\Delta_{1,1} \mid \Phi_{3,2}; \Delta_{1,1} \mid \Psi_{3,2}$
22	3R	Γ_4	$(\Delta_3 \Gamma_4) (\Xi_3 \Delta_3 \Gamma_4)$	<i>need</i> $(\gamma^-, 1 : \beta^+)$; avail $\Gamma_{4,3}$	$\Gamma_{3,3} \mid \Delta_{3,1}; \Gamma_{3,3} \mid \Lambda_{3,1}$
23	3R	Γ_4	$(\Pi_3 \Delta_3 \Gamma_4) (\Xi_3 \Delta_3 \Gamma_4)$	<i>need</i> $(\gamma^-, 1 : \beta^+)$; avail $\Gamma_{4,3}$	$\Xi_{3,2} \mid \Delta_{3,1}; \Xi_{3,2} \mid \Lambda_{3,1}$
24	3R	Γ_4	$(\Psi_3 \Delta_3 \Gamma_4) (\Xi_3 \Delta_3 \Gamma_4)$	<i>need</i> $(\gamma^-, 1 : \beta^+)$; avail $\Gamma_{4,3}$	$\Psi_{3,2} \mid \Delta_{3,1}; \Psi_{3,2} \mid \Lambda_{3,1}$
25	1R	Γ_4	$(\Sigma_4 \Gamma_3) (\Delta_3 \Gamma_4)$	<i>need</i> $(\gamma^-, 1 : \beta^+)$; avail $\Gamma_{4,3}$	$\Sigma_{4,3} \mid \Gamma_{3,3}$
26	7R	Γ_5	$(\Sigma_4 \Gamma_3) (\Pi_3 \Gamma_5 \Phi_1)$	<i>need</i> $(\alpha^+, 1 : \beta^+)$; avail $\Gamma_{5,3}$	$\Gamma_{4,3} \mid \Gamma_{2,2}; \Lambda_{4,3} \mid \Gamma_{2,2}$
27	4R	Γ_5	$(\Pi_3 \Sigma_3 \Delta_1) (\Psi_3 \Gamma_5 \Phi_1)$	<i>need</i> $(\alpha^+, 1 : \beta^+)$; avail $\Gamma_{5,3}$	$\Phi_{5,2} \mid \Psi_{2,2}; \Phi_{5,2} \mid \Xi_{2,2}$
28	4R	Γ_5	$(\Psi_3 \Sigma_3 \Delta_1) (\Psi_3 \Gamma_5 \Phi_1)$	<i>need</i> $(\alpha^+, 1 : \beta^+)$; avail $\Gamma_{5,3}$	$\Psi_{5,2} \mid \Psi_{2,2}; \Psi_{5,2} \mid \Xi_{2,2}$
29	2R	Γ_5	$(\Gamma_0 \Sigma_1) (\Psi_3 \Gamma_5 \Phi_1)$	<i>need</i> $(\alpha^+, 1 : \beta^+)$; avail $\Gamma_{5,3}$	$\Theta_{1,1} \mid \Pi_{2,2}; \Xi_{1,1} \mid \Pi_{2,2}$
30	6R	Σ_0	$(\Phi_4 \Sigma_5 \Phi_0) (\Delta_4 \Sigma_0 \Pi_0)$	<i>need</i> $(\zeta^+, 1 : \theta^-)$; avail $\Sigma_{0,3}$	$\Delta_{3,3} \mid \Delta_{2,3}; \Delta_{3,3} \mid \Sigma_{2,3}$
31	4R	Σ_0	$(\Phi_4 \Gamma_2 \Psi_0) (\Delta_4 \Sigma_0 \Pi_0)$	<i>need</i> $(\zeta^+, 1 : \theta^-)$; avail $\Sigma_{0,3}$	$\Psi_{5,4} \mid \Phi_{2,4}; \Psi_{5,4} \mid \Theta_{2,4}$
32	4R	Σ_0	$(\Phi_4 \Gamma_2 \Psi_0) (\Delta_4 \Sigma_0 \Psi_0)$	<i>need</i> $(\zeta^+, 1 : \theta^-)$; avail $\Sigma_{0,3}$	$\Psi_{5,4} \mid \Psi_{2,4}; \Psi_{5,4} \mid \Xi_{2,4}$

Complete grouped classification (3 of 3)

#	N	Break	Post-image pattern	Required / available	Representative pre-images
33	35R	Σ_1	$(\Delta_4 \Sigma_0 \Pi_0) (\Gamma_0 \Sigma_1)$	<i>need</i> $(\delta^+, 1 : \theta^-)$; avail $\Sigma_{1,3}, \Sigma_{1,5}$	$\Delta_{2,3} \mid \Gamma_{1,1}; \Delta_{2,3} \mid \Lambda_{1,1}$
34	14R	Σ_1	$(\Gamma_1 \Delta_0 \Pi_0) (\Gamma_0 \Sigma_1)$	<i>need</i> $(\delta^+, 1 : \theta^-)$; avail $\Sigma_{1,3}, \Sigma_{1,5}$	$\Theta_{0,5} \mid \Gamma_{1,1}; \Theta_{0,5} \mid \Lambda_{1,1}$
35	7R	Σ_1	$(\Phi_4 \Gamma_2 \Pi_0) (\Gamma_0 \Sigma_1)$	<i>need</i> $(\delta^+, 1 : \theta^-)$; avail $\Sigma_{1,3}, \Sigma_{1,5}$	$\Gamma_{5,3} \mid \Gamma_{1,1}; \Gamma_{5,3} \mid \Lambda_{1,1}$
36	1R	Σ_1	$(\Gamma_1 \Delta_0) (\Gamma_0 \Sigma_1)$	<i>need</i> $(\delta^+, 1 : \theta^-)$; avail $\Sigma_{1,3}, \Sigma_{1,5}$	$\Gamma_{0,3} \mid \Sigma_{1,3}$
37	1R	Σ_1	$(\Phi_4 \Sigma_5 \Lambda_0) (\Gamma_0 \Sigma_1)$	<i>need</i> $(\delta^+, 1 : \theta^-)$; avail $\Sigma_{1,3}, \Sigma_{1,5}$	$\Sigma_{3,3} \mid \Delta_{1,1}$
38	6R	Σ_2	$(\Phi_3 \Sigma_2 \Phi_1) (\Phi_3 \Sigma_2 \Phi_1)$	<i>need</i> $(\gamma^-, 1 : \beta^+)$; avail $\Sigma_{2,3}$	$\Delta_{0,3} \mid \Pi_{0,1}; \Delta_{0,3} \mid \Psi_{0,1}$
39	6R	Σ_2	$(\Pi_3 \Sigma_3 \Delta_1) (\Phi_3 \Sigma_2 \Phi_1)$	<i>need</i> $(\gamma^-, 1 : \beta^+)$; avail $\Sigma_{2,3}$	$\Delta_{5,2} \mid \Delta_{0,3}; \Delta_{5,2} \mid \Lambda_{0,3}$
40	3R	Σ_2	$(\Gamma_0 \Sigma_1) (\Phi_3 \Sigma_2 \Phi_1)$	<i>need</i> $(\gamma^-, 1 : \beta^+)$; avail $\Sigma_{2,3}$	$\Gamma_{1,1} \mid \Delta_{0,3}; \Gamma_{1,1} \mid \Lambda_{0,3}$
41	3R	Σ_2	$(\Phi_3 \Sigma_2 \Phi_1) (\Phi_3 \Sigma_2 \Theta_1)$	<i>need</i> $(\gamma^-, 1 : \beta^+)$; avail $\Sigma_{2,3}$	$\Delta_{0,3} \mid \Gamma_{0,1}; \Lambda_{0,3} \mid \Gamma_{0,1}$
42	2R	Σ_2	$(\Lambda_3 \Sigma_2 \Phi_1) (\Phi_3 \Sigma_2 \Phi_1)$	<i>need</i> $(\gamma^-, 1 : \beta^+)$; avail $\Sigma_{2,3}$	$\Sigma_{0,3} \mid \Pi_{0,1}; \Sigma_{0,3} \mid \Psi_{0,1}$
43	2R	Σ_2	$(\Pi_3 \Gamma_5 \Phi_1) (\Phi_3 \Sigma_2 \Phi_1)$	<i>need</i> $(\gamma^-, 1 : \beta^+)$; avail $\Sigma_{2,3}$	$\Gamma_{2,2} \mid \Pi_{0,1}; \Gamma_{2,2} \mid \Psi_{0,1}$
44	1R	Σ_2	$(\Lambda_3 \Sigma_2 \Phi_1) (\Phi_3 \Sigma_2 \Theta_1)$	<i>need</i> $(\gamma^-, 1 : \beta^+)$; avail $\Sigma_{2,3}$	$\Sigma_{0,3} \mid \Gamma_{0,1}$
45	1R	Σ_2	$(\Pi_3 \Gamma_5 \Phi_1) (\Phi_3 \Sigma_2 \Theta_1)$	<i>need</i> $(\gamma^-, 1 : \beta^+)$; avail $\Sigma_{2,3}$	$\Gamma_{2,2} \mid \Gamma_{0,1}$
46	1R	Σ_2	$(\Sigma_4 \Gamma_3) (\Lambda_3 \Sigma_2 \Phi_1)$	<i>need</i> $(\gamma^-, 1 : \beta^+)$; avail $\Sigma_{2,3}$	$\Delta_{4,3} \mid \Sigma_{0,3}$
47	35R	Σ_3	$(\Sigma_4 \Gamma_3) (\Pi_3 \Sigma_3 \Delta_1)$	<i>need</i> $(\alpha^+, 1 : \beta^+)$; avail $\Sigma_{3,3}$	$\Gamma_{4,3} \mid \Delta_{5,2}; \Gamma_{4,3} \mid \Lambda_{5,2}$
48	18R	Σ_3	$(\Gamma_0 \Sigma_1) (\Psi_3 \Sigma_3 \Delta_1)$	<i>need</i> $(\alpha^+, 1 : \beta^+)$; avail $\Sigma_{3,3}$	$\Delta_{1,1} \mid \Pi_{5,2}; \Delta_{1,1} \mid \Psi_{5,2}$